WEEK 05: Generating Functions

1 Exercises

Exercise 1. Find the coefficient of:

a) x^{10} in $(1 + x + x^2 + x^3 + \dots)^n$
b) x^r in $(x^5 + x^6 + x^7 + \dots)^8$
c) x^7 in $(1 + x^2 + x^4)(1 + x)^m$
d) x^{16} in $(x + x^2 + x^3 + x^4 + x^5)(x^2 + x^3 + x^4 + \dots)^5$
e) x^{17} in $(x^2 + x^3 + x^4 + x^5 + x^6 + x^7)^3$

Exercise 2. Find the coefficient of x^{11} in:

a) $x^2(1 - x)^{-10}$
b) $\frac{x^2-3x}{(1-x)^4}$
c) $\frac{(1-x^2)^5}{(1-x)^5}$

Exercise 3. Use generating functions to find the number of ways to select 10 balls from a large pile of red, white, and blue balls, given that:

a) The selection must include at least two balls of each color.
b) The selection has at most two red balls.
c) The number of blue balls selected is an even number.

Exercise 4. Use generating functions to find the number of ways to distribute r jelly beans to eight children if:

a) Each child receives at least one jelly bean.
b) Each child receives an even number of jelly beans.

Exercise 5. How many ways are there to distribute 20 coins to n children and one adult, given that the adult receives either 5 coins or 10 coins and:

a) Each child can receive any number of coins?
b) Each child receives at most 1 coin?

Exercise 6. How many ways can 300 chocolate candies be selected from seven different types, given that each type is packed in boxes of 20 and one must select at least one box but no more than five boxes of each type?

Exercise 7. Prove that $(1 - x - x^2 - x^3 - x^4 - x^5 - x^6)^{-1}$ is the generating function for the number of ways a sum of r can occur when a die is rolled any number of times.

Solution 17:

Exercise 8. Construct a generating function for a_r, the number of ways to distribute r identical objects into:

a) Five different boxes, each containing at most three objects.
b) Three different boxes, each containing between three and six objects.
c) Six different boxes, each containing at least one object.
d) Three different boxes, where the first box contains at most five objects.

Exercise 9. Find the generating function for a_r, the number of ways to distribute r identical objects into n different boxes, such that the first box contains an *odd* number of objects between r_1 and s_1, the second box contains an *even* number of objects between r_2 and s_2, and the remaining boxes contain at most three objects.

Exercise 10. Use generating functions to model the number of ways to choose five integers from $\{1, 2, \ldots, n\}$ such that no two integers are consecutive.